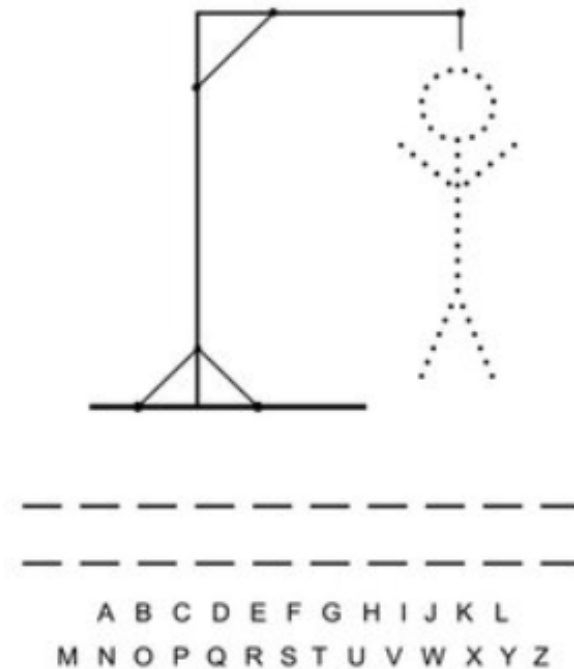


Hangman!



Name: Eder Jorvin Guerrero

Department: Computer Science and Information Engineering

Course Name: Introduction to Computer Programming II

School: National Dong Hwa University

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Introduction

Hangman is a word-guessing game where one player thinks of a word and the other player tries to guess it by suggesting letters. The word is represented by a series of blank spaces, each corresponding to a letter in the word. A part of a stick figure (the hangman) is drawn for each incorrect letter guessed. If a correct letter is chosen, all the blank spaces belonging to that letter will be filled. The game is won if the guesser identifies the word before completing the hangman drawing.

Conversely, the game is lost if the entire hangman is drawn before the word is guessed. In this version of the game, the player's word will be chosen at random from a database. The player will also be given the definition of the word that is to be guessed. The game also give the player an option to create their own set of words.

Technologies Utilized

IDE:

- Visual Studio Community 2022
 - Use [medium.com...link](#) in the references to view how to configure visual studio community 2022 to connect to the database.

Database:

- MySQL workbench 8.0 CE
- `mysql-connector-c++-8.4.0-winx64`

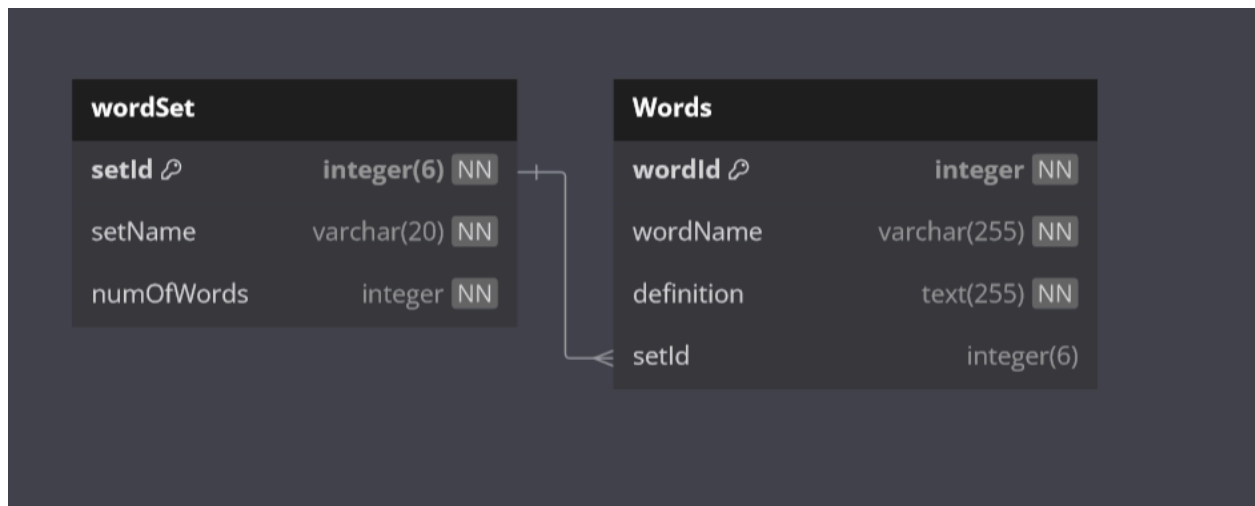
Drawing the schema:

- [dbdiagram.io](#)
 - The link can be found in the references.

Database

The game uses a relational database named hangman with two tables: wordSet and Words. It has a one to many relationship. Navigate to additionalResources folder. It will contain two files hangman.sql and database_scheme_diagram_code.txt. Use hangman.sql to create the database used in the hangman game. The database will only contain one set of words name computer science. However, the images below contain the changes made while testing. Use database_schema_diagram_code to view the schema of the database. The link to insert the code is found in the file.

Database schema:



wordSet Table:

	setId	setName	numOfWords
▶	1	computer science	4
	3	test	2
	4	test2	3
	5	test3	1
★	NULL	NULL	NULL

Words Table:

	wordId	wordName	definition	setID
▶	1	Computer Science	The scientific study of computation, applied to b...	1
	2	complexity	The way that a solution to a problem scales as t...	1
	3	Browser	A piece of software that enables a user to locat...	1
	4	Function	A small section of computational code that perf...	1
	5	eder	my name	3
	6	idk	this is a test	3
	7	hello	it is a greeting	4
	8	bye	it is a farewell	4
	9	dog	an extremely loved animal	4
	10	hi	one of the most common greetings	5
★	NULL	NULL	NULL	NULL

Features

Create a new set of words

- The system will allow the user to create their own set of words to use in the hangman game.
- The user will be prompted to name the set and then enter a list of words and the definition of each word.

UI:

- The user is provided with a text-base user interface.
- The user is presented with different options and a number to execute the option.

About:

- The user is given a description of the game.

Gameplay

- When the user selects to start the game, they are prompted with a different menu displaying the list of set of words currently saved. If the user has not created his own set of words, it will only contain the default set.

- Once the user has selected the set of words that he wants to use for the game, the system will generate a random number and use the random number to select a word from the list.
- The game view will be displayed and the user will be prompted to start guessing.
- The system has a section to display the letters already chosen and does not let the user to choose the same letter more than once.
- If the user selects a wrong letter, the image of the hangman is updated.
- At the end of the game, the user is given a statement with the result of the game and then is send back to the main menu.

References

<https://medium.com/@neel.lad3110/featuring-mysql-database-in-visual-studio-2022-for-a-c-application-a90befbc0420>

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